

BioLink

AI-Powered Drug Repurposing Discovery

Turning a research model into a tool that patients, researchers, and clinicians can actually use.

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LIVE DEMO

huggingface.co/spaces/dustinhaggett/biolink-v2

Drug development is broken.

Thousands of existing drugs could help — if we could find the connections.

10–15

YEARS

to bring a new drug from discovery to market

\$2.6B

PER DRUG

average cost per approved drug (DiMasi et al., 2016)

>90%

FAILURE

of drug candidates never make it through trials

Repurposing has worked before.

Thalidomide · sedative → multiple myeloma treatment

Minoxidil · blood-pressure med → hair-loss treatment

Metformin · diabetes drug → cancer & aging research

THE GAP

No accessible tool combines ML predictions with real-world evidence for drug repurposing.

Manual process. Deep domain expertise required.

Access barrier. Existing databases are expensive and specialized.

Why AI?

The drug-disease relationship space is too large for humans to search manually.

18M possible drug-disease pairs · **380K** known associations in CTD · **0.947** AUC on held-out data

LEVEL 1

Prediction

MLP Classifier

Trained on ~380K CTD pairs. Scores all 7,163 drugs in ~100 ms.

Test AUC 0.947

LEVEL 2

Understanding

Claude Haiku

Resolves natural-language input to CTD entities and generates plain-English explanations.

~\$0.001 per query

LEVEL 3

Validation

Perplexity Sonar

Searches published literature in real time to ground every prediction in evidence.

5 APIs integrated

What BioLink Does

A web app that makes drug repurposing research accessible to everyone.

FORWARD SEARCH

Search by Disease

"What drugs might treat Lyme disease?"

Ranks all 7,163 drugs with calibrated confidence — Strong / Moderate / Speculative.

REVERSE SEARCH

Search by Drug

"I'm taking Metformin — what else might it help?"

Scores all 2,525 diseases against a single drug using the same evidence pipeline.

PLUS

Evidence Verdicts

Supports · Conflicts · Standard-of-Care · Insufficient

Clinical Trials

Active trial counts +
ClinicalTrials.gov links

Compare & Export

Side-by-side tables · PDF reports ·
CSV export

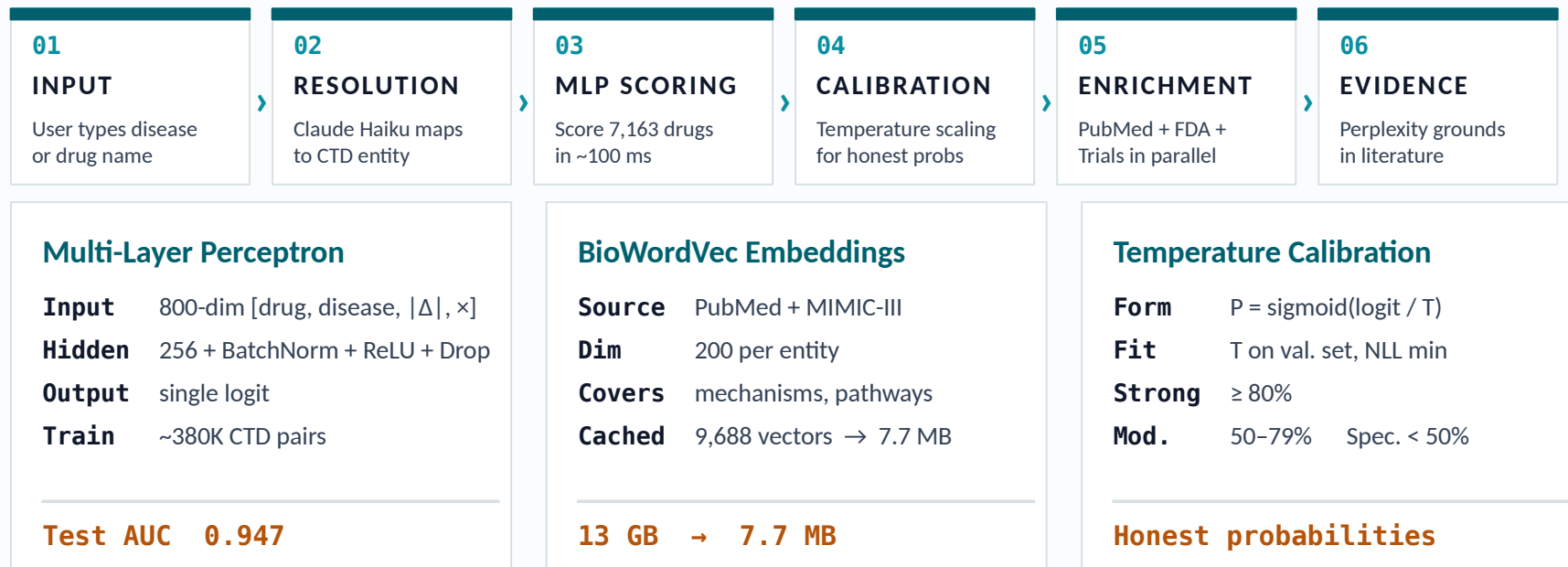
Safety First

Drug interactions · evidence-quality ratings · disclaimers

DEPLOYED · huggingface.co/spaces/dustinhaggett/biolink-v2 — works without API keys; keys unlock full LLM features.

System Architecture

A 6-stage pipeline from natural-language input to evidence-grounded results.



Multi-API Pipeline

Five APIs working together — three free, two paid (under \$0.01 per query).

ENGINE	PRIMARY FUNCTION	RESPONSE	COST
Claude Haiku (Anthropic)	Entity Resolution & Explanations	Structured JSON	~\$0.001
Perplexity Sonar	Evidence Search & Follow-up Q&A	Structured JSON	~\$0.005
PubMed E-utilities (NCBI)	Drug-Disease Co-occurrence Counts	XML / JSON	FREE
OpenFDA	Drug Approval Status Lookup	JSON	FREE
ClinicalTrials.gov (v2)	Active & Completed Trial Finder	JSON	FREE

All enrichment runs in parallel via Python **asyncio** — minimal added latency to the pipeline.

When the model disagrees with the evidence.

The most valuable feature is showing users when the model is wrong.

CYCLOSPORINE

CONFLICTS

Model: **99% Strong** for Lyme disease

Evidence: **CONFLICTS**

Cyclosporine is an immunosuppressant. Suppressing immunity during a bacterial infection would be harmful. The model detects a real relationship — but can't distinguish *'interacts'* from *'treats'*.

DOXYCYCLINE

STANDARD-OF-CARE

Model: **99% Strong** for Lyme disease

Evidence: **STANDARD-OF-CARE**

Already the first-line antibiotic treatment. Multiple RCTs and CDC guidelines confirm. Model and evidence align perfectly — this is the actionable result.

Key insight. The gap between model confidence and real-world evidence **IS** the information.

Feature Set

Every feature designed around one question: "Can a patient actually use this?"

● Evidence Verdicts

Color-coded badges: literature supports, conflicts, or is insufficient.

● Evidence Quality

RCT › Human Study › Preclinical
› Case Report › Theoretical.

● Clinical Trials

Live ClinicalTrials.gov lookup with counts and enrollment links.

● Pathway Chains

Visualizes mechanism: drug → target → pathway → effect.

● Compare Mode

Select multiple candidates for side-by-side metric comparison.

● Follow-up Q&A

Ask questions about any result, grounded in published evidence.

● Drug Interactions

Warns when a candidate interacts dangerously with standard care.

● PDF / CSV / Batch

Doctor-ready reports. Bulk predictions via CSV upload.

Each feature exists to convert raw model output into a decision someone can defend.

Built for three audiences.

Same engine. Different surfaces. The interface earns trust before it makes claims.

PATIENT

I want answers I can bring to my doctor.

- Plain-English explanations
- Confidence tiers, not raw scores
- PDF reports for clinic visits
- Follow-up Q&A on any result

RESEARCHER

I want to screen many candidates fast.

- Batch CSV upload (5 diseases / run)
- PubMed co-occurrence counts
- ClinicalTrials.gov links per row
- CSV export for downstream analysis

CLINICIAN

I want to compare safely and quickly.

- Side-by-side compare mode
- Drug-interaction warnings
- Evidence-quality classification
- Pathway visualization

DESIGN PRINCIPLE • Manrope + Inter typography, teal-on-white minimalism, progressive disclosure. *Disclaimers top and bottom of every screen.*

What we hit. What fixed it.

Key technical obstacles and how AI tooling helped resolve them.

Model vs. Evidence Mismatch	MLP gives high confidence to drugs that evidence shows are harmful.	→ Perplexity evidence search with verdict badges (agree / disagree).
API Cost at Scale	Perplexity at $\sim \$0.01/\text{call} \times 20 \text{ results} = \$0.20/\text{query}$ — cost-prohibitive.	→ Evidence-enrich top 5; free APIs (PubMed, FDA, Trials) for all 20.
Free-Text Entity Resolution	"Heart attack" needs to map to "Myocardial Infarction" (CTD term).	→ Claude Haiku NL → entity mapping with difflib fuzzy fallback.
13 GB Deployment Size	BioWordVec binary too large for HF Spaces (50 GB total cap).	→ Pre-cache only the 9,688 vectors actually needed → 7.7 MB.

0.947 TEST AUC

~100ms INFERENCE

7.7 MB DEPLOYMENT

AI was the product and the build tool.

We used AI both at runtime and as our primary development environment.

BUILD - TIME

Development tooling

Claude Code (Anthropic) — primary tool for architecture, code generation, debugging, and documentation across all 10 modules.

GitHub Copilot — code completion during early development.

→ 10-module, ~2,000-line app with 5 API integrations — built in one semester.

RUN - TIME

Inside the product

Claude Haiku (Anthropic) — entity resolution (NL → CTD term) and plain-English explanations. ~\$0.001/query

Perplexity Sonar — live evidence search; structured output (verdict, quality, pathway, citations). ~\$0.005/query

AI tools collapsed the timeline from concept to deployed application into a single semester.

Repurposing as a force multiplier.

Where this matters most: the populations big pharma can't profitably serve.

95% of rare diseases have no FDA-approved treatment. Repurposing is often the only viable path.

● Rare Diseases

7,000+ rare diseases affect 300M people. Novel drugs aren't profitable for small populations.

● Health Equity

Repurposed generics are dramatically cheaper. The discovery process itself is free here.

● Pandemic Readiness

Repurposing screens identified dexamethasone for COVID-19 in months, not 10–15 years.

● Patient Empowerment

Patients with unmet needs can bring evidence-backed PDF reports to their physicians.

RESPONSIBLE AI DESIGN

BioLink does not replace clinical judgment. Every screen carries disclaimers. Evidence verdicts surface where the AI is wrong. The tool is a hypothesis generator — not a treatment decision-maker.

Demo flow.

Six steps. Disease in, evidence-grounded ranked candidates out.

LIVE huggingface.co/spaces/dustinhaggett/biolink-v2

1 Search Disease

Enter "Lyme disease" — get ranked candidates with confidence scores.

2 Review Tiers

Strong ($\geq 80\%$), Moderate (50–79%), Speculative ($< 50\%$).

3 Evidence Verdicts

Compare Doxycycline (Standard-of-Care) vs Cyclosporine (Conflicts).

4 Clinical Detail

Expand pathway chains and ClinicalTrials.gov enrollment links.

5 Compare Mode

Select two candidates for side-by-side metric comparison.

6 Reverse Search

Switch to drug mode — search "Metformin" for candidate diseases.

Thank you.

Questions & discussion.

LINKS

GitHub github.com/dustinhaggett/biolink-v2

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TEAM

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Kera Prosper

Mia Braun

BioLink is a hypothesis-generation tool for research purposes only. Not medical advice. Consult a healthcare provider.